

A method on mapping the distribution of photovoltaic power stations in complex terrain regions

Hongzhi Mao^{a,b}, Weili Liu^a, Xie Chen^{a,c}, Zhiyong Tian^{a,*}, Angelo Zarrella^b,
Yongqiang Luo^a, Jianhua Fan^c, Wentao Wu^d

^a Hubei Key Laboratory of Multi-media Pollution Cooperative Control in Yangtze Basin, School of Environmental Science & Engineering, Huazhong University of Science and Technology, 1037 Luoyu Road, Wuhan, Hubei, 430074, China

^b Department of Industrial Engineering, University of Padova, Via Venezia 1, 35131, Italy

^c Department of Civil and Mechanical Engineering, Technical University of Denmark, Brovej 118, 2800, Kgs. Lyngby, Denmark

^d Department of Civil and Natural Resources Engineering, University of Canterbury, Christchurch, New Zealand

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ABSTRACT

With the goal of achieving carbon neutrality, the number of new photovoltaic (PV) installations worldwide has surged in recent years. Accurate statistics on the area and distribution of centralized solar PV installations are essential. Due to land use constraints and poverty alleviation policies, an increasing number of PV stations are being installed in mountainous areas in China. The fragmented distribution of these PV stations presents challenges for remote sensing identification. This study proposes a method based on Object-Based Image Analysis (OBIA) and the Cascade Random Forest Classifier, utilizing Sentinel-1/2 imagery. The method is capable of extracting PV stations across diverse terrains, including mountains, plateaus, and plains. Specifically, five different scenarios with varying feature combinations (including spectral, index, geometric, and texture features from Sentinel-1 and Sentinel-2) were compared. A second-level random forest classifier, trained on false-positive samples, was employed to enhance the model's robustness. The results indicate that the spectral features of Sentinel-1, along with index and geometric features, positively influence identification accuracy, while texture features showed less impact. Independent test results demonstrated an overall classification accuracy of 99.9904 %, with a producer accuracy of 92.73 % and a user accuracy of 81.38 % for PV stations in a complete city region. Finally, a 2024 map of centralized PV installations was created for five cities west of Beijing, where the proportion of total PV area in mountainous regions is 45 %. This method represents the first approach capable of identifying PV stations in complete regions with complex terrain, including mountains, without the need for manual post-processing.

1. Introduction

Carbon neutrality is a global consensus. As of July 2024, 148 countries have established carbon neutrality development goals [1]. Solar photovoltaic (PV) has become one of the most important renewable energy technologies for achieving carbon neutrality and addressing energy crises, with installed capacity growing rapidly. In 2023, the new global PV installations increased by 444 GW, which was 176 % of the last year's. Among them, China added 216.3 GW, becoming the largest driver of global PV growth [2]. The United States [3], Germany [4], Spain [5], and several other countries also set new records for the new PV installations. Notably, in the U.S. [3], it accounted for more than half

of all new electricity-generating capacity additions in 2023 for the first time. Meanwhile, global policies promoting renewable energy are constantly being strengthened. The European Union has increased its target for the share of renewable energy in total energy consumption by 2030 from 30 % to 42.5 % [6]. China has set a goal of exceeding 450 GW of the new PV power generation by 2024. Therefore, rapid growth in PV installations is anticipated in the future. On the other hand, it is noteworthy that in China, due to the increasing restrictions on the occupation of cropland and forests and the PV poverty alleviation project, mountainous and hilly areas have become the most suitable and popular installation sites in the Beijing-Tianjin-Hebei region [7]. This strategy alleviates development imbalances in this region, and also reduces the

* Corresponding author.

E-mail addresses: tianzy0913@163.com, zhiyongtian@hust.edu.cn (Z. Tian).

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cost of long-distance power transmission from PV stations in northwest China [8]. Moreover, large rooftop PV stations on industrial buildings are becoming common [9].

It is of significance to estimate the total installed PV capacity and distribution accurately. This can facilitate a better understanding of the progress towards carbon neutrality goals, evaluate the effectiveness of government incentive policies [10], monitor the encroachment of PV on cropland and forest land [11], and analyze the impact of PV stations on the local microclimate [12]. Studies have shown that large-scale centralized PV stations in deserts increase local humidity and promote desert degradation [13]. Smaller-scale centralized PV stations can reduce the amplitude of the seasonal cycle of local temperatures [14]. Lower global carbon emissions in future will also increase the efficiency of PV power generation [15,16].

Only a few countries have complete data of the locations and scales of PV installations, and such data are typically not publicly accessible. Using publicly available satellite remote sensing imagery to identify the centralized PV stations has become a popular method recently. Ishii et al. [17] are the first team to use Landsat 8 imagery. They identified centralized PV stations in Japan with a capacity greater than 5 MW. Subsequently, Zhang et al. [18] used six spectral bands and three index features, along with texture and terrain features, to train a random forest (RF) model that mapped the centralized PV distributions in Ningxia Province, Northwest China. They noted that the index and texture features can improve identification accuracy. They further used spectral and index features to map centralized PV distributions across China [19]. Chen et al. [20] and Wang et al. [21] used seven bands from the Landsat 9 satellite to identify centralized PV distributions in the Golmud region and the Gansu Province, Northwest China, respectively. These studies demonstrated that Landsat satellites could achieve high identification accuracy for large-scale, contiguous centralized PV stations, especially in the deserts, gobi regions. However, the distribution of centralized PV installed in mountainous and hilly terrain is fragmented and Landsat imagery can't perform well in identification in these regions, due to the limitation of 30m/px resolution. Some high-precision satellite images such as SPOT [22], and Gaofen [23] are also used for this work, however, they are not easily available publicly for free.

The recently launched Sentinel 1/2 platform, with higher resolutions (10–20m) and free public access, can help alleviate the issues faced by Landsat. Jiang et al. [24] utilized spectral and index features calculated from Sentinel-1 and Sentinel-2 to develop a RF classifier based on Pixel Based Image Analysis (PBIA), creating a 10m resolution centralized PV map of China's coastal provinces. Feng et al. used Sentinel-2 imagery and the PBIA method to map centralized PV installations across China in 2020 [25]. Due to the large identification range, which spans multiple satellite swaths, the issue of different spectra for the same objects in different swaths was addressed by establishing separate training sets and training corresponding RF classifiers for each province. And due to the salt-and-pepper noise, the PBIA had to be accompanied by pixel quantity threshold filtering and multiple morphological post-processing steps [26]. These resulted in a significant loss of the fragmented PV areas in mountainous and hilly regions. Furthermore, the results obtained from this method still require manual visual checking and filtering, which entails substantial labor costs.

Object-Based Image Analysis (OBIA) can effectively mitigate the issues faced by PBIA. This method integrates adjacent pixels with similar spectral features into an object, segmenting the complete image into numerous objects. The classifier is then trained and to identify these objects. Ladisa et al. [27] used Sentinel-2 imagery and OBIA to identify centralized PV stations in a region of Lazio, Italy. Tao et al. [28] and Stid et al. [29] utilized OBIA to map centralized PV distributions in Massachusetts and California's Central Valley, respectively. Xia et al. [30] analyzed spectral features, index features, and the geometric features of objects to identify centralized PV stations in five provinces of northwest China. Plakman et al. [31] trained a RF classifier using Sentinel-1 and Sentinel-2 imagery in the Netherlands. Their training, validation, and

test sets were sourced from different provinces and satellite swaths. To maintain the classifier's accuracy in unfamiliar cities, they randomly altered the brightness of PV objects in the training set as a form of data augmentation. However, in current researches, the OBIA method to identify PV stations has only been applied to flat plains and plateaus terrains, mainly involving cropland, grassland, desert, and Gobi. There is a lack of studies on complex terrains such as mountainous and hilly areas where PV distributions are highly fragmented. It is also worthwhile to optimize methods that can be identified in unfamiliar satellite swath regions.

Deep learning has also gradually been used in this work. Deep learning refers to a class of machine learning techniques that use deep neural networks to automatically learn and extract patterns from large datasets. These models, particularly Convolutional Neural Networks (CNNs), are designed to handle high-dimensional data and are commonly applied in tasks like image classification and semantic segmentation. Various forms of CNN models have been proposed, including Candidate Region-based Methods [32], Fully Convolutional Network (FCN) [33], and weakly supervised methods [34], and so on. Kruitwagen et al. [22] utilized a cascade classifier consisting of two U-Nets to identify large-scale centralized PV stations in the world. Most deep learning models involve image downsampling and resolution reduction steps, which can degrade the precision and accuracy of PV boundary delineation. In complex terrains, PV power stations are distributed as small size blocks with irregular shapes, being highly sensitive to the model's contour identification capabilities. Currently, there is a lack of accurate high-resolution datasets for PV in complex terrains.

The above analysis indicates that existing methods face challenges in accurately identifying centralized PV stations in complex terrains, such as mountains and hills, at high resolution. There is a notable lack of accurate distribution and profile datasets for centralized PV stations in these areas. This deficiency leads to an insufficient quantity and quality of training sets for deep learning models, which hinders their performance in PV identification in complex terrains.

To address this issue and obtain accurate high-resolution distribution data for centralized PV stations in complex terrains, based on Sentinel imagery and OBIA, a method is proposed for identifying centralized PV distributions, which can be applied to various complex terrains automatically without manual post-processing. It can have high identification accuracy in unfamiliar new regions. Different features and combinations have been discussed and optimized. Furthermore, in a region of China that includes diverse terrains such as mountains, plateaus, and plains, a map with high-resolution and accurate centralized PV distributions has been established.

2. Materials and methods

The workflow for identification PV stations is shown in Fig. 1. It mainly includes three steps: image preprocessing, image segmentation, and classifier training. Image preprocessing involves a sequence of steps, including image selection, cloud removal, mean calculation, and color correction (histogram matching). Images were collected and pre-processed in Google Earth Engine (GEE), then downloaded to local. Later, the images are input into eCognition software for segmentation using the OBIA method. Image segmentation results in the generation of numerous objects. By calculating the shape, spectral, index, and texture features of these objects, a vector dataset is created, with different feature combinations depending on the scenario.

Some objects are labeled based on auxiliary data, with categories including PV and non-PV objects. These objects are then divided into training, validation, and test sets based on their category and quantity. The training set is used to train the RF classifier. The RF algorithm is chosen because it has been proven for RF to have a higher identification accuracy compared with other machine learning methods [35]. The classifier was then applied to two validation sets from different cities to analyze the impact of different feature scenarios on identification

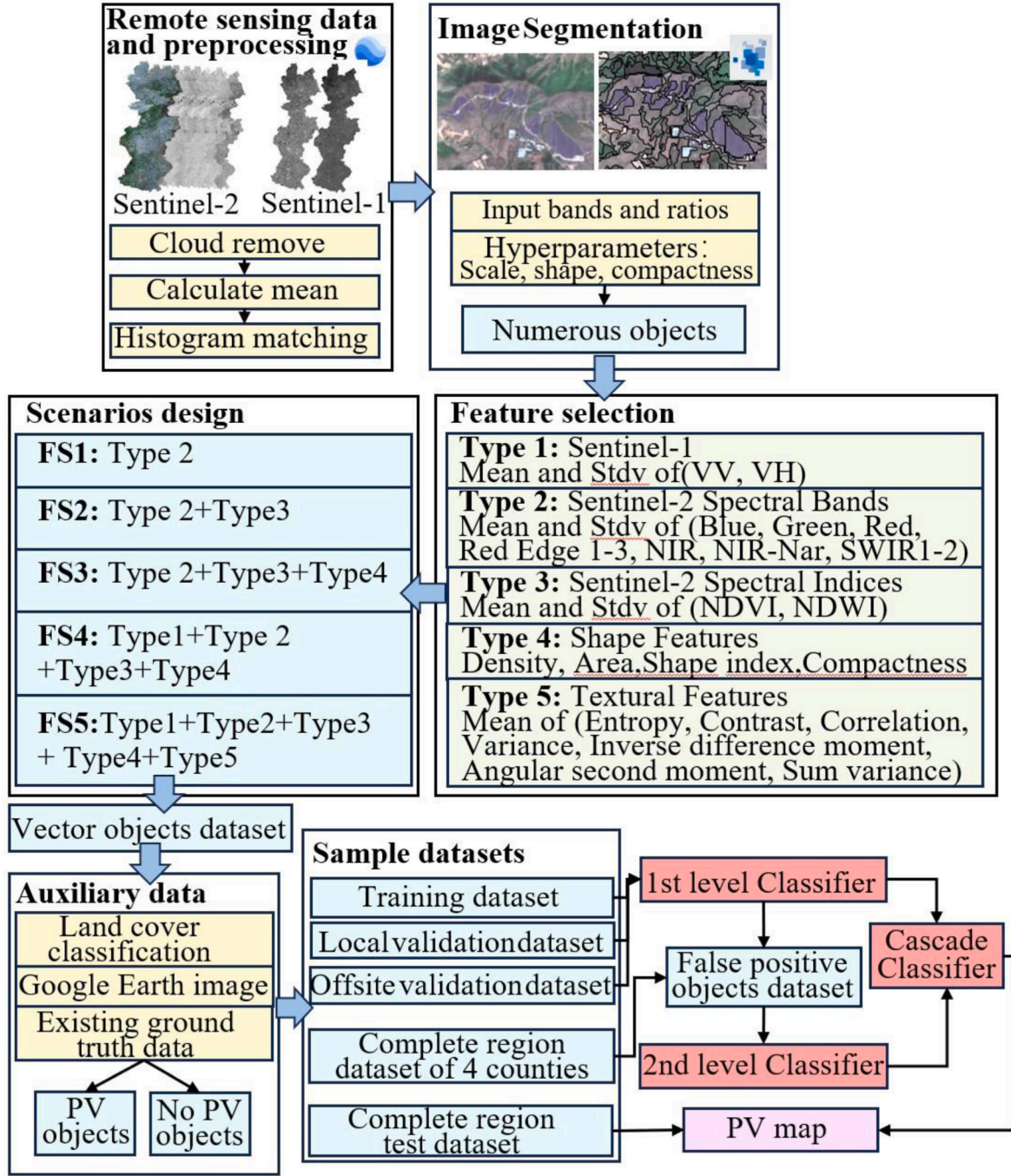


Fig. 1. Workflow for identification PV stations in complex terrain region in this study.

accuracy. Feature selection was performed based on feature importance for optimization. The optimized classifier was used to identify the complete regions of 4 counties, and the false positive samples results, as supplementary datasets, were used to train a second-level RF classifier. The two RF classifiers together formed a cascading classifier.

2.1. Study area

The study area includes five cities located in the Taihang Mountains in Hebei province, China, as shown in Fig. 2. The Beijing-Tianjin-Hebei region is the third most economically developed city cluster in China. It has a large population and a high demand for electricity. Meanwhile, the rural areas in the mountainous lag in development, with limited arable land. It is one of the key regions in China to carry out PV poverty

alleviation policy recently. Hebei is the province with the largest amount of centralized PV installations in China 2023, at 54.16 GW, widely distributed in the western mountains and the northern plateau [36].

From south to north, the Taihang Mountains cross along the western part of Hebei province, while the northern part belongs to the Bashang Plateau. The central and southern parts belong to the North China Plain.

Study area, Western Hebei five cities, Zhangjiakou, Baoding, Shijiazhuang, Xingtai, and Handan, were selected as the areas for establishing the training set and the first validation set. This region covers a total area of 96300 km².

Two other cities were selected as validation and test regions. Yangquan, located adjacent to the western part of Hebei, was chosen as the area for the second validation set, covering a total area of 4559 km².

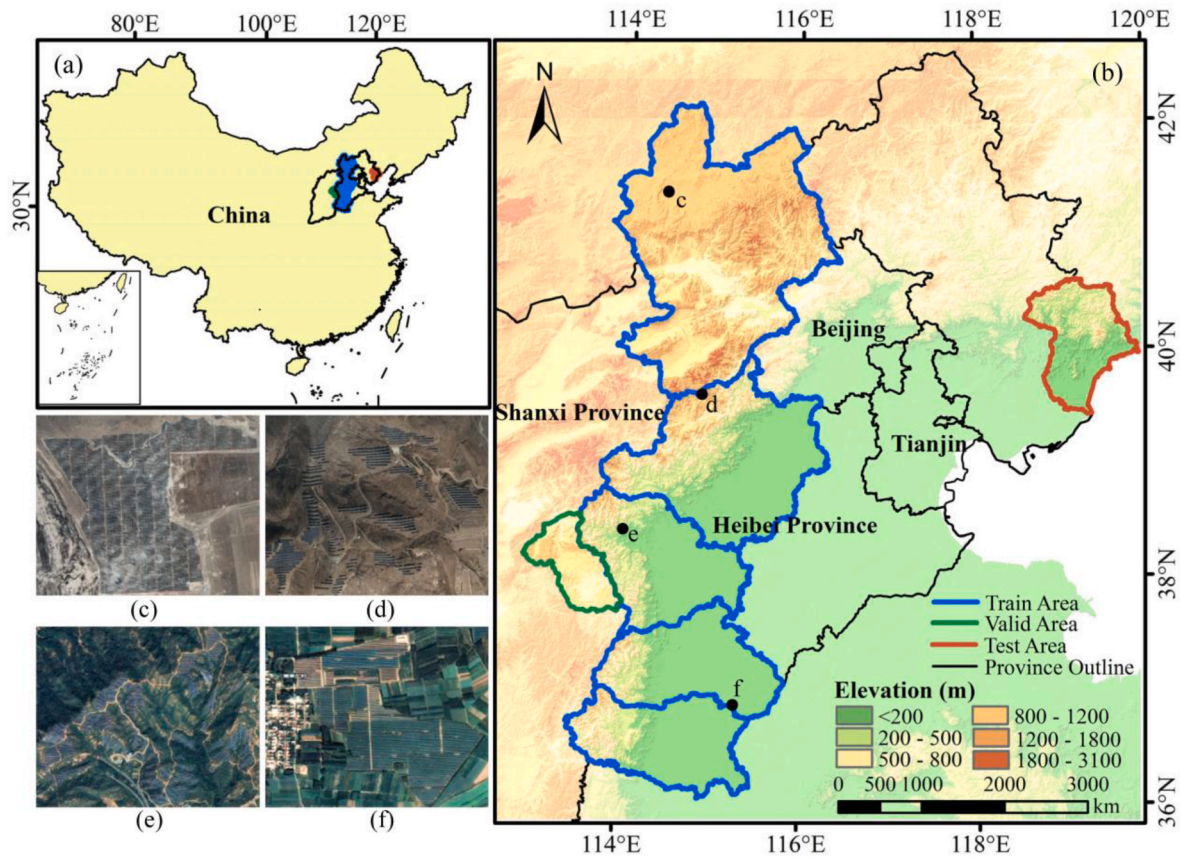


Fig. 2. Geographical location of the study area. (a) Geographic location of the train, valid, test area in China. (b) Elevation distribution in the study area. (c) Aerial image of a PV station in plateaus; (d) and (e) Aerial images of two PV stations in mountains; (f) Aerial image of a PV station in Plains.

Qinhuangdao, situated in the northeastern part of Hebei, was selected as the test region, with a total area of 7802 km².

2.2. Image preprocessing

Sentinel-2 and Sentinel-1 remote sensing imagery is used. All images were loaded and preprocessed from the GEE platform. Sentinel-2 provides multispectral imagery with 13 bands. Here, we utilized 10 bands ranging from visible light to shortwave infrared, including four 10m/px resolution bands and six 20m/px resolution bands, as shown in Table 1. We specifically selected images with less than 1 % cloud coverage between June 1 and September 30, 2023, and their number distribution is shown in Fig. 3. After cloud removal, all images were superimposed and the average of every pixel were calculated.

The Sentinel-1 spacecraft is equipped with a C-band synthetic aperture radar (SAR). Due to the longer wavelength of the C-band, some

Table 1

Description of the remote sensing data used in this study.

Sensor used	Bands	Wavelength	Resolution
Sentinel-2	B2 (Blue)	496.6 nm (S2A)/492.1 nm (S2B)	10
	B3 (Green)	560 nm (S2A)/559 nm (S2B)	10
	B4 (Red)	664.5 nm (S2A)/665 nm (S2B)	10
	B5 (RE1)	703.9 nm (S2A)/703.8 nm (S2B)	20
	B6 (RE2)	740.2 nm (S2A)/739.1 nm (S2B)	20
	B7 (RE3)	782.5 nm (S2A)/779.7 nm (S2B)	20
	B8 (NIR)	835.1 nm (S2A)/833 nm (S2B)	10
	B8A (RE4)	864.8 nm (S2A)/964 nm (S2B)	20
	B11 (SWIR 1)	1613.7 nm (S2A)/1610.4 nm (S2B)	20
	B12 (SWIR 2)	2202.4 nm (S2A)/2185.7 nm (S2B)	20
Sentinel-1	VV	5.405 GHz	10
	VH	5.405 GHz	10

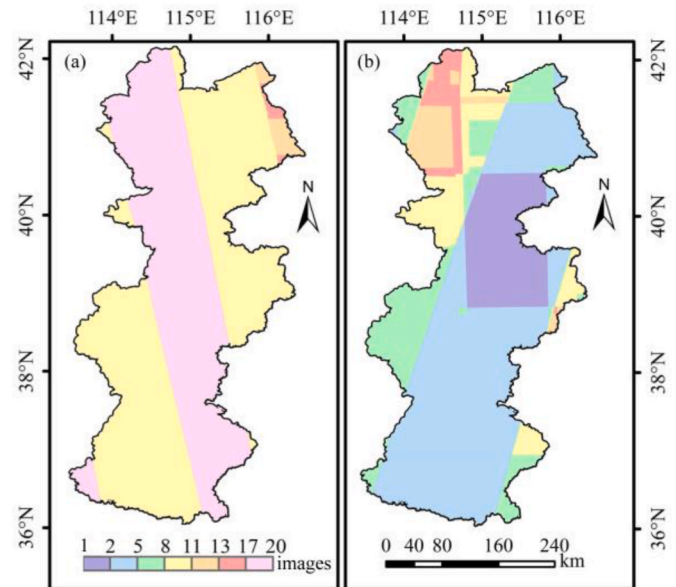


Fig. 3. Statistics of the training set region remote sensing data in 2023. (a) Sentinel-1 data, (b) Sentinel-2 data.

electromagnetic waves can penetrate clouds, allowing SAR imagery to overcome cloud interference. Here, we selected Sentinel-1 VV and VH 10m/px images from June 1 to September 30, 2023 from GEE.

As the frost-free period in the northern part of Hebei Province is less than 120 days, this timeframe effectively minimizes the impact of snow

and frost. When applying this method to other regions, it is important to select images from periods that avoid months with a high likelihood of snow and frost.

As shown in Fig. 3, the images corresponding to the study areas are

distributed in different swaths. Therefore, even though the average value of every pixel over time was calculated, there may still be spectral diversity for the same class. In this study, all images underwent color correction. We compared histogram matching and normalization

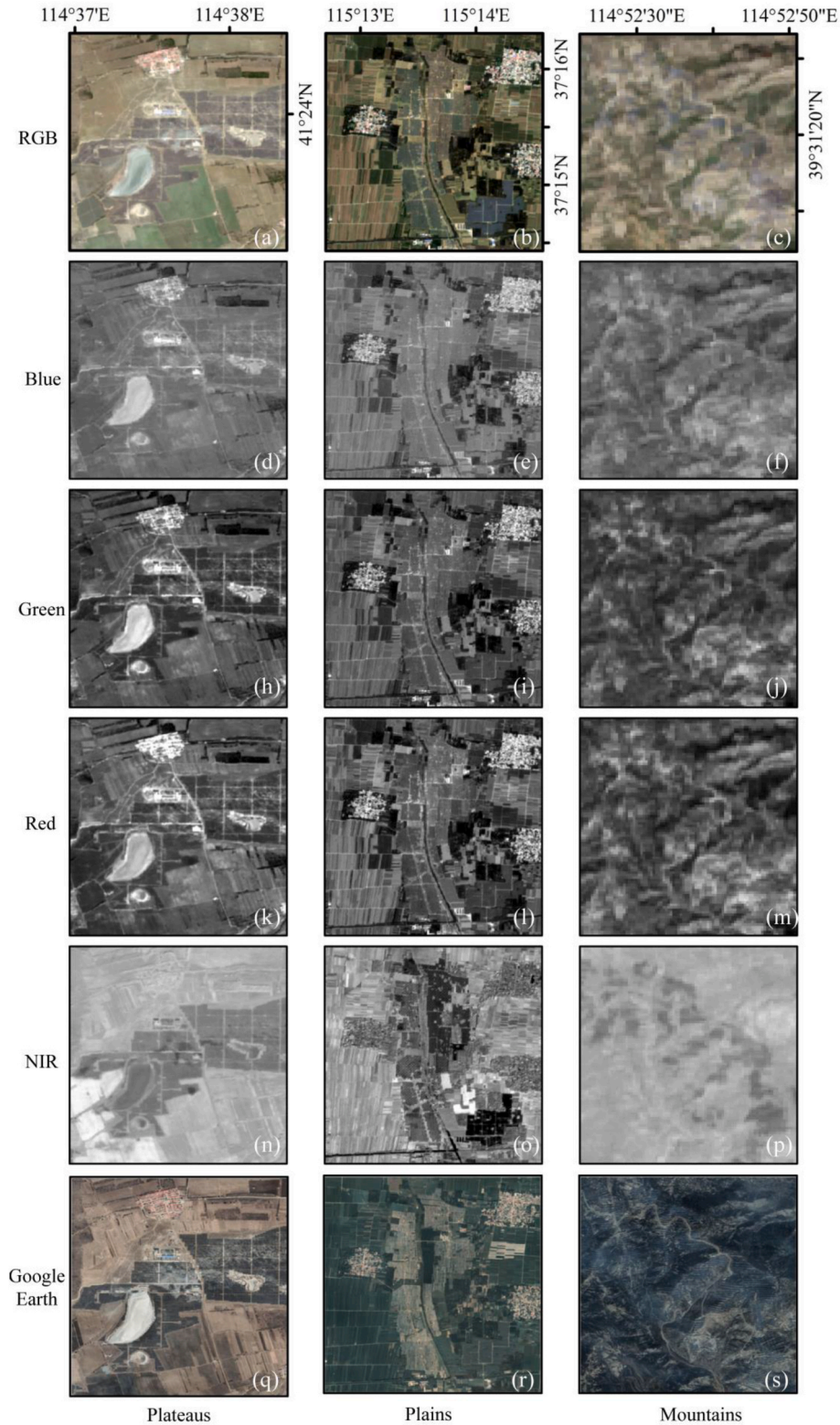


Fig. 4. Difference in reflectance between PV stations and surroundings in different terrains under visible and NIR bands with 10m resolution. (a)~(c) RGB images synthesized by sentinel-2 in different terrains. (d)~(f) Images in the Blue band. (h)~(j) Images in the Green band. (k)~(m) Images in the Red band. (n)~(p) Images in the NIR band. (q)~(s) Aerial images from Google Earth. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

methods, combining them with subsequent feature scenarios and RF training.

2.3. OBIA image segmentation

After preprocessing, the images were further image segmented to generate a large number of homogenized objects. The eCognition software is utilized for multi-scale segmentation. The hyperparameters of multi-scale segmentation include scale, shape factor and compactness factor, and the number and weights of spectral bands involved in the segmentation. Scale denotes the lower limit of the size of segmentation objects. The shape factor indicates the emphasis on shape versus spectrum during segmentation. The compactness factor affects the extensibility of the objects.

The visible and near-infrared (NIR) bands of Sentinel-2 have higher resolution and can show the boundaries of PV stations more clearly. Therefore, these four bands were initially chosen as inputs for image segmentation. Among them, single-band images are compared, revealing that PV stations exhibit significantly lower reflectance than their surrounding environment in the NIR band, resulting in clear boundary demarcation, as shown in Fig. 4. This reflectance difference is not apparent in the other three bands. Hence, increasing the proportion of the NIR band may enhance the accuracy of the objects of PV station boundaries. Additionally, considering that many studies have shown the B11 band to be the most important for identifying PV [31], the B11 band was selected. Various proportions were compared, and it was found that $B2:B3:B4:B8:B11 = 1:1:1:1.75:0.25$ resulted in the clearest PV boundaries.

The study area includes mountainous regions, plains, and plateaus. To find a set of hyperparameters that can simultaneously satisfy the segmentation requirements for different terrains, various scenarios with different hyperparameters were compared. Through manual visual judgment, the optimal hyperparameters were determined to be 70-0.7-0.5 for plateaus, 30-0.8-0.5 for plains, and 50-0.8-0.5 for mountainous regions. The hyperparameters of 50-0.7-0.5 were chosen, which is the best set of parameters to take into account the segmentation effect of each terrain. At these hyperparameters in Fig. 5(d)–(f), it can be seen

that the shape of the objects changes with the landcover in a way, that fits more closely to the boundary of the landcover. In Fig. 5(e), at the northeast corner of the water area, the objects have a larger scale.

2.4. Ground truth and establishment of sample sets

Through referencing the centralized PV public databases supported by Zhang et al. [19] and Wang et al. [37], the approximate locations of most centralized PV stations can be obtained. And with high-resolution imagery from Google Earth, objects located at the edges of PV stations and small, fragmented PV areas can be accurately labeled. Notably, due to the distinct low reflectance of PV station areas in the NIR band, the combination of the B8 band and RGB images can greatly facilitate judgment and labeling.

A binary classification approach was used for the labeling work, distinguishing only between PV and no-PV samples. To enhance sample diversity, no-PV samples were selected based on the land cover categories of the WorldCover v100 global map. An equal number of samples were randomly collected from each category. The study area comprises seven land use types, including tree cover, grassland, cropland, bare/sparse vegetation, permanent water bodies, herbaceous wetlands, and built-up. Considering the potential confusion of PV stations with roads, glass roofs, and agricultural greenhouses [38], the number of built-up type objects was appropriately increased.

Ultimately, a sample set containing 9594 objects was formed in the five cities of the training set region. Among them, 80 % were extracted as the training set, and 20 % as the first validation set, also named the local validation set. In the Yangquan, 1918 objects were generated to serve as the second validation set, also named the offsite validation set. The complete Qinhuangdao area was used as the test set, containing 188 PV objects and 489865 no-PV objects. The object numbers of each sample set are shown in Table 2.

The quantity and diversity of no-PV samples in the training set may not be insufficient, potentially leading to more false positive results in the first-level RF classifier used for complete region identification. To train the second-level RF classifier, a part of the false positive results identified by the first-level RF classifier in the 4 complete regions are

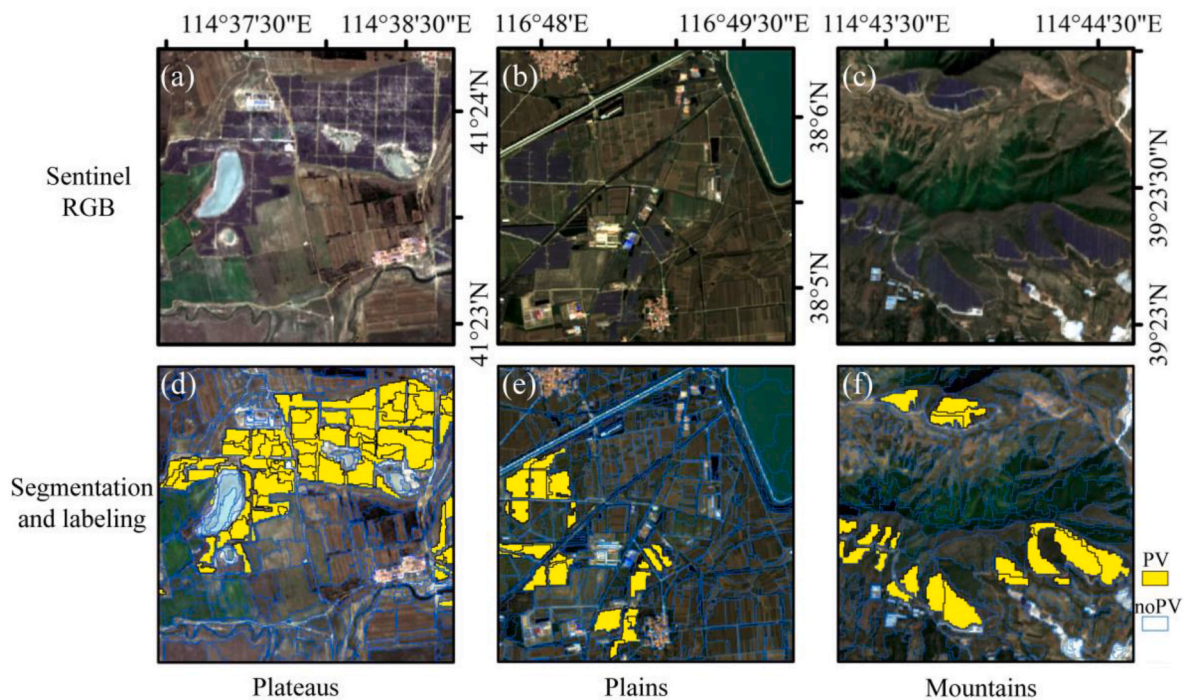


Fig. 5. Segmentation results and labeling of PV stations and their surroundings in different terrains. (a)–(c) RGB images synthesized by sentinel-2 in different terrains. (d)–(f) objects in images segmented by multiscale segmentation.

Table 2

The number and total area of objects in the training, validation and test datasets.

Set	Category	Objects number	Total Area (km ²)
Training dataset	Photovoltaic	3838	98.15
	No Photovoltaic	3838	94.34
Validation dataset-1	Photovoltaic	959	25.48
	No Photovoltaic	959	22.57
Validation dataset-2	Photovoltaic	994	15.96
	No Photovoltaic	994	23.09
Test dataset	Photovoltaic	188	2.77
	No Photovoltaic	489865	75.25

used as no-PV samples in the training set for the second-level RF classifier. These 4 counties are Xuanhua District in Zhangjiakou, Laiyuan County and Lianchi District in Baoding, and Xinji County in Shijiazhuang. The numbers of objects in these areas were 116444, 120943, 17887, and 50440, respectively. They represent land covers of mountainous and basin regions, mountainous regions, plain cities, and plain rural regions.

2.5. Feature selection and scenario establishment

4 types of features were used for classification in this study: spectral features, index features, geometric features, and texture features. First, the spectral features included the mean and standard deviation of Sentinel satellite bands. For each object, the features were calculated based on the mean and standard deviation of all pixel values within this object.

Second, the index features included the mean and standard deviation of the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Water Index (NDWI), calculated as Eq. (1) and Eq. (2) [39,40]. These indexes are effective in filtering out vegetation and water bodies.

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (1)$$

$$NDWI = \frac{Green - NIR}{Green + NIR} \quad (2)$$

Third, as unique features to OBIA, geometric features were chosen. Geometric features include object area, density, shape index and compactness index, calculated by Eq. (3) ~ Eq. (5).

$$Shape\ index = \frac{P}{2\sqrt{\pi \times Area}} \quad (3)$$

$$Density = \frac{Area}{Width \times Height} \quad (4)$$

$$Compactness = \frac{Width \times Height}{Area} \quad (5)$$

where the P is the perimeter of the object, $Width$ and $Height$ are the bounding rectangular box of the object.

Lastly, texture features were calculated from the Gray Level Co-occurrence Matrix (GLCM) from a grayscale image obtained by the weighted average of the NIR and SWIR1 bands, including seven characteristics as shown in Fig. 1. The grayscale image is calculated as given in Equation (6). The effects of texture features with window sizes of 2, 4, 8, and 16 were compared.

$$Gray = 0.5NIR + 0.5SWIR1 \quad (6)$$

To compare the performance of different features, five feature scenarios were designed and computed. FS1 included only the spectral features of Sentinel-2. FS2 to FS4 sequentially added index features, geometric features, and Sentinel-1 spectral features to compare the effect of introducing each type of feature on model improvement. FS5

included Sentinel-2 spectral features, index features, geometric features, and texture features.

2.6. RF model and evaluation metrics

In this study, we tested the effect of different number of RF trees on the identification accuracy, including 100,200, 300,400 and 500. Finally, 500 trees were chosen because it has the highest accuracy.

Based on the best color correction method, each scenario was extracted and the corresponding first-level RF classifier was trained. After determining the best feature scenario on accuracy in validation sets, the importance of each feature was calculated, and the features with low importance were gradually deleted and retrain the classifier again. The feature streamlining process is run until there is a significant decrease in the F1measure value.

The optimal RF classifier was ultimately applied to the complete area of four counties to collect identified false-positive samples. Based on these false-positive samples, the second-level RF classifier was trained. Finally, the filtering algorithm of distance threshold within 200 m was used.

For evaluating the model, the confusion matrixes calculated from the validation and test sets were used to determine the producer accuracy (PA), user accuracy (UA), and F1-measure of the PV.

3. Results

3.1. Comparison of different image color correction

Table 3 compares the identification results of the first-level RF classifier trained under three different image color correction with FS3 scenario. It can be observed that on the local validation set, the results obtained from the different color correction methods show no significant differences. On the offsite validation set, the PA of PV obtained through histogram matching is 88.32 %, significantly higher than normalization and raw images. It demonstrates that image color correction is necessary when the model is applied to areas outside the training set swatches. Moreover, the histogram matching method proves to be superior.

3.2. Comparison of different feature scenarios

Table 4 compares the identification results of the first-level RF classifier using texture features with 4 different window sizes. It is shown that the size 4 achieves the best F1-measure on the local validation set at 0.9525. On the offsite validation set, a size of 2 achieves the best F1-measure, with a value of 0.8615, but it has the lowest PA at 80.76 %. In contrast, a size of 4 has the best PA on both validation sets, with values of 95.42 % and 81.85 %, respectively.

The results of the first-level RF classifiers with 5 feature scenarios are compared in Table 5. By comparing the F1-measure, FS4 achieves the highest accuracy on both the local and offsite validation sets. The UA and PA of FS4 on the local validation set are 96.12 % and 95.52 %, respectively, while on the offsite validation set in Yangquan, the UA and PA are 92.27 % and 88.32 %, respectively. From FS1 to FS4, as the types of features increase, the F1-measure also increases. It indicates that

Table 3

Comparison of the effect of different color corrections on the validation sets.

Dataset	Raw Image	Normalization	Histogram matching
Val1-UA	94.74 %	96.66 %	96.12 %
Val1-PA	95.45 %	94.97 %	95.52 %
Val1-F1	95.09 %	95.81 %	95.82 %
Val2-UA	91.56 %	98.55 %	92.27 %
Val2-PA	67.95 %	37.11 %	88.32 %
Val2-F1	78.01 %	53.92 %	90.25 %

Table 4

Comparison of validation sets metrics from different texture feature window sizes.

Metrics	FS5-S2	FS5-S4	FS5-S8	FS5-S16
Val1-UA	95.60	95.08	95.07	95.07
Val1-PA	94.16	95.42	93.94	93.83
Val1-F1	94.87	95.25	94.50	94.45
Val2-UA	92.32	90.56	91.14	90.16
Val2-PA	80.76	81.85	81.22	81.48
Val2-F1	86.15	85.98	85.89	85.60

Table 5

Comparison of validation sets metrics from five different feature scenarios.

Metrics	FS1	FS2	FS3	FS4	FS5-S4
Val1-UA	95.38	95.71	95.92	96.12	95.08
Val1-PA	94.78	95.51	95.62	95.52	95.42
Val1-F1	95.08	95.61	95.77	95.82	95.25
Val2-UA	91.23	91.88	90.91	92.27	90.56
Val2-PA	86.92	86.62	87.53	88.32	81.85
Val2-F1	89.02	89.17	89.19	90.25	85.98

index features, geometric features, and Sentinel-1 spectral features all positively contribute to the PV identification. FS5 has the lowest F1-measure, indicating that the introduction of texture features does not improve the model's identification accuracy.

Overall, the combination of spectral, index, and geometric features of Sentinel-1&2 is the best feature scenario, regardless of whether the classifier is applied to a local or offsite application set.

3.3. Optimization of the number of features

Fig. 6(a) calculates the importance of 32 features in the RF model in the FS4 scenario. The four most important features are B11_mean, VH_stdDev, NDWI_mean, and B12_mean. Overall, the importance of the standard deviation features of most spectral bands is lower than that of the mean features of the corresponding bands.

To improve the computational efficiency, features with low importance were gradually removed. It can be seen from Fig. 7 that as the number of features decreases, the F1-measure on the validation sets initially decreases and then slowly increases, eventually then exceeding the result of the original 32 features. Until the number of features is reduced to 11, the F1-measure on the offsite validation set reaches its highest at 90.86 % (UA = 89.23, PA = 92.56 %). At this point, the local validation set also reaches its highest PA at 96.45 %, with a UA of 96.15 %. When the number of features is further reduced to 10, the identification accuracy on the offsite validation set begins to decline.

Therefore, the model with 11 features 11 was chosen as the best first-level RF classifier. Its feature importance ranking is shown in Fig. 6(b). It includes the mean values of B2, B3, B4, B5, B8A, B11 and B12 bands, the mean value of NDWI, the mean and standard deviation of VH bands, and the area of objects. The mean of B11 and the mean of B12 are the two most important features, which is consistent with many other studies [30,31]. The B8A band is retained and the B8 band is removed, and the wavelength range of the former is included in that of the latter. This shows that the Narrow NIR band (B8A) is more representative of the spectral feature of centralized PV stations than the NIR band (B8).

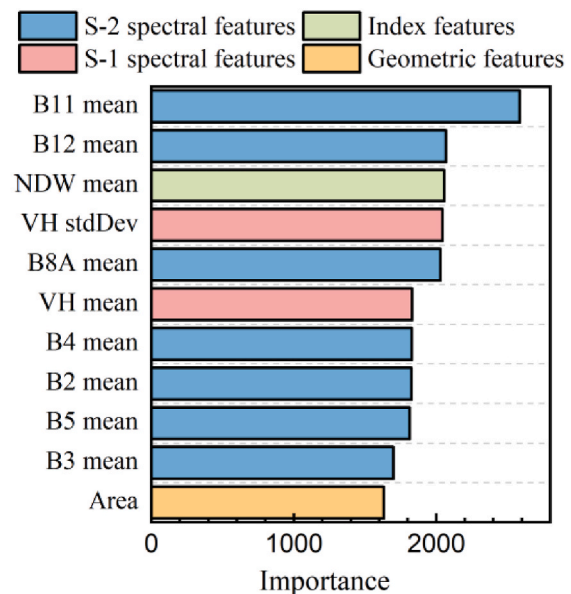
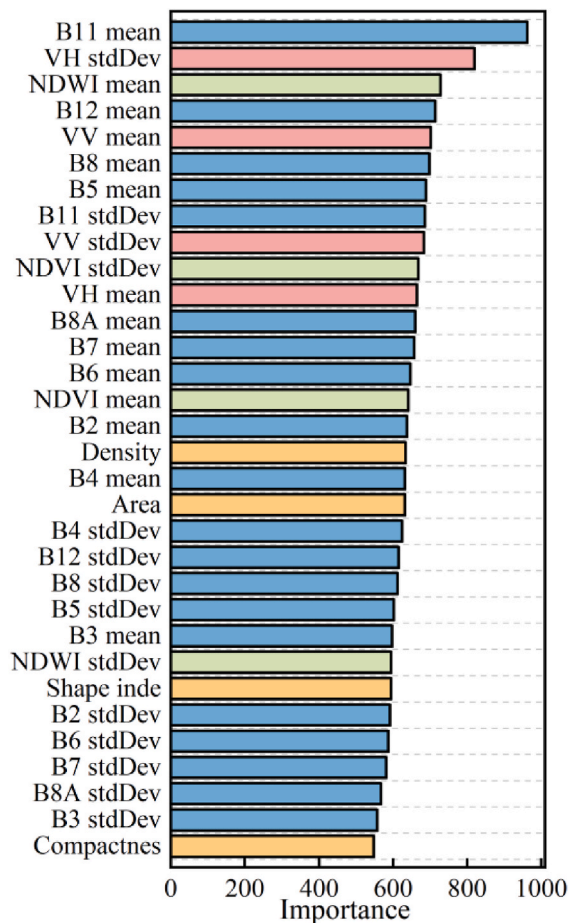


Fig. 6. Importance ranking of features in RF classifiers. (a) Importance ranking of all 32 features in FS4 feature scenario, (b) Importance ranking of features in streamlined best classifiers of FS4.

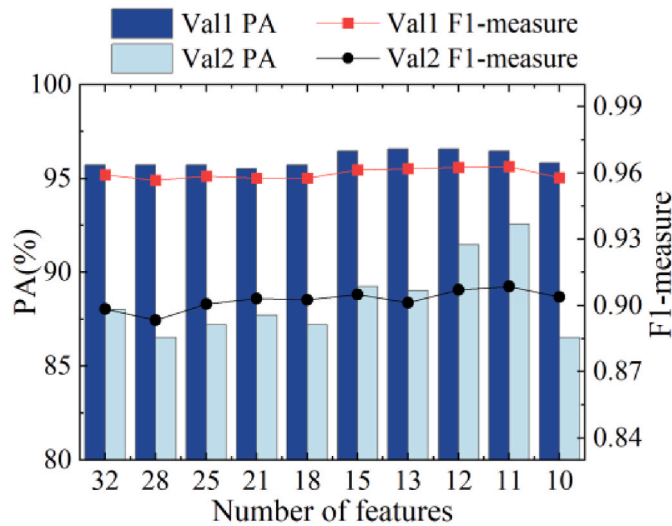


Fig. 7. Changes in the relationship between the number of features and model accuracy during the streamlining process.

3.4. Establishment of the cascade classifier

The optimal first-level RF classifier was applied to the complete regions of four counties. The results yielded 5128, 720, 291, and 385 false positive samples, respectively, as shown in Table 6. The average false positive rate is 1.85 %. The highest false positive rates are observed in the Xuanhua with rates of 4.4 %, representing mountains.

To establish a supplementary training set, maintaining a balance among different terrain categories to some extent. In the first round, 5 % of the false positive samples from Xuanhua and Laiyuan, 20 % from Lianchi and Xinji, and 15 % of the no-PV samples from the original training set were randomly selected, along with an equal number of PV samples. After applying the cascade classifier to four counties again, the average false positive rate dropped to 0.41 %. Then, all false positive samples from the second round were mixed with the previous supplementary training set, to retrain the second-level RF classifier. The results showed only 9 false positive objects in the four counties.

Then the cascade classifier was applied again to the validation sets, the resulting confusion matrices and accuracy metrics are shown in Table 7. On the local validation set, the PV UA was 89.89 %, PV PA was 98.07 %, and PV F1-measure was 0.9380. On the offsite validation set, the PV UA was 83.00 %, PV PA was 99.52 %, and PV F1-measure was 0.9051.

3.5. Model testing

The cascade classifier was applied to the test set. The results, shown in Table 8, indicate that the first-level RF classifier identified 172 true positive samples while generating 10,859 false positive samples. After the filtering of the second-level classifier and the distance threshold

Table 6
Process of building the supplementary training set.

Supplementary training region	xuanhua	laiyuan	lianchi	xinji
Objects number of entire region	116444	120943	17887	50440
Number of FP by the first model	5128	720	291	385
FP rate by the first model	4.40 %	0.60 %	1.63 %	0.76 %
Number used to supplementary training dataset	256	36	58	77
Number of FP by the 1st supplementary	929	720	31	39
FP rate after the 1st supplementary	0.80 %	0.60 %	0.17 %	0.08 %
Numer used to supplementary training dataset	465	360	31	39
Number of FP after the 2nd supplementary	4	4	1	0
FP rate after the 2nd supplementary	0.003 %	0.003 %	0.006 %	0
Total number of supplementary objects	721	396	89	116

Table 7

Confusion matrix computed by the cascade classifier on the validation sets.

Dataset and classifier		Ground truth		UA and F1
		No PV	PV	
Val1-1st-level classifier	No	921	38	UA (no PV) = 96.04 %
	PV	36	923	
Val1-2nd-level classifier	No	21	17	UA (PV) = 96.24 % F1(PV) = 0.9380
	PV	61	862	
Val1-PA		90.66 %	98.07 %	
Val2-1st-level classifier	No	1027	100	UA (no PV) = 99.65 %
	PV	76	918	
Val2-2nd-level classifier	No	96	4	UA (PV) = 83.00 % F1 (PV) = 0.9051
	PV	93	825	
Val2-PA		86.92 %	99.52 %	

The map categories are the rows and the reference categories are the columns.

Table 8

Confusion matrix computed by the cascade classifier on the test set.

cascade classifier		Ground truth		UA, F1 and OA
		No PV	PV	
Test dataset 1st-level classifier	No	479006	10859	UA(no PV) = 99.998 % UA(PV) = 81.38 % F1(PV) = 0.8669 OA = 99.9904 %
	PV	16	172	
Test dataset 2nd-level classifier	No	10716	143	
	PV	17	155	
After distance filter	No	131	12	
	PV	2	153	
Final total PA		489888	153	
		99.99 %	92.73 %	

The map categories are the rows and the reference categories are the columns.

filtering, there are 153 true positive objects and 12 false positive objects left, with UA = 81.38 % and PA = 92.73 % for PV on the complete test set. The overall accuracy for this region was 99.9904 %. The PV stations map in test set is shown in Fig. 8.

The test sets in most studies are selected randomly, the overall accuracies of the model may cannot represent the results when applied in a complete region. In this study, the method can obtain 99.9904 % of the overall accuracy on Qinhuangdao, a complete region with a total area of 7803 km². Without other post-processing and manual processing, we can get high accuracy PV prediction results and more complete and clear PV boundaries.

3.6. Map the PV distribution in 2024

Through this method, the updated 2024 centralized PV distribution

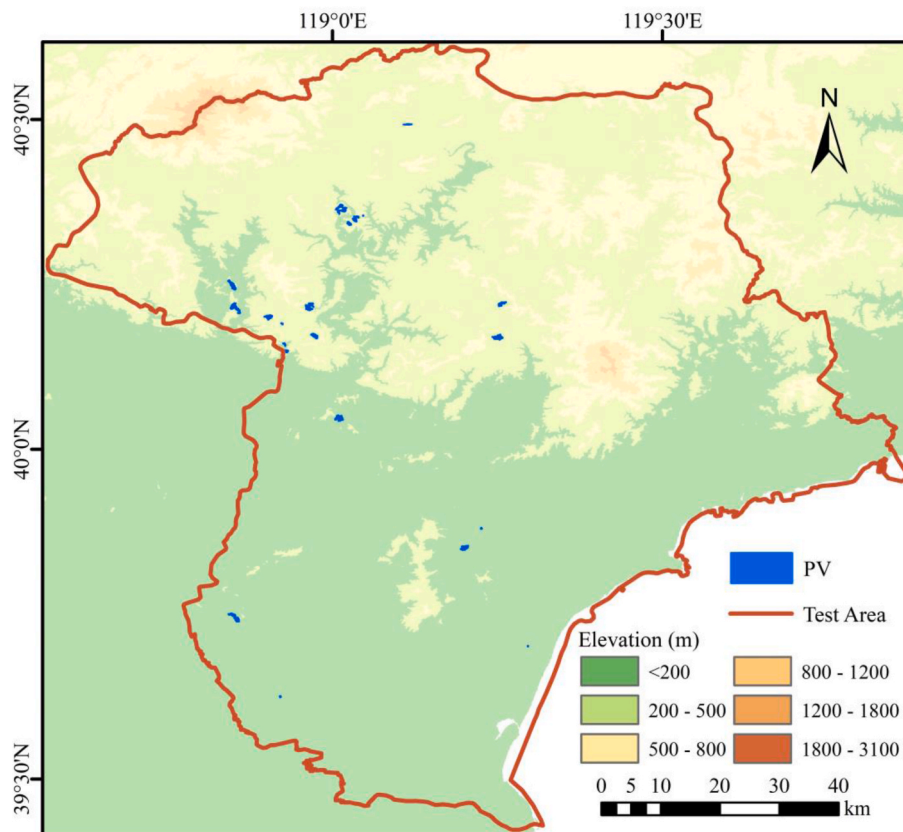


Fig. 8. PV stations map in test set of Qinhuangdao complete region.

maps for five cities in the western Hebei Province were established, as shown in Fig. 9. A total of 256 km² of PV area was identified. In addition to centralized PV, many large distributed PV were also accurately identified. Among them, the PV area distributed in the northern plateau terrain is 110.9 km², the total PV area in the central and western mountainous is 116.2 km², while the total area in the eastern plains is 29.8 km². Fig. 9 (b)–(d) show the PV in three different terrains respectively, and they are not collected by other public datasets up until now. As shown, the method can accurately identify the PV contours under different terrains. The total PV area in mountainous accounts for about 45 % of these five cities. It suggests that the accurate identification of PV in mountains and other complex terrains is important for total statistical data.

3.7. Electricity generation estimation

The list of PV poverty alleviation projects, published by the Chinese Ministry of Finance, provides information on the installed capacities and locations of numerous PV stations, with detailed data at the village level [41]. Using the village locations, corresponding satellite images were downloaded, and the associated centralized PV stations were identified. A total of 267 centralized PV stations with each capacity exceeding 10 MW were successfully identified and matched. Fig. 10 illustrates the relationship between the area and the installed capacity of these PV stations. Based on these findings, the total installed capacity of centralized PV stations in the study area is estimated to be 12,885 MW.

Long-term yearly average potential PV electricity production (PVO_{UT}), measured in kWh/kWp, is provided by Solargis and the World Bank Group for East China, covering the period from 2007 to 2018 [42]. Fig. 11(a) depicts the distribution of PVO_{UT} across the study area. Using this data, the PVO_{UT} for each centralized PV station within the study area can be determined. The annual electricity generation for each county has been estimated separately, as shown in Fig. 11(b). The total

annual electricity generation for the 98 counties across five cities in the study area is projected to be 2.0614 billion kWh. Among these, Zhangbei County in the northern region is expected to generate the highest output, with an estimated 5.349 billion kWh.

Currently, the grid-connected electricity price for PV in Hebei Province is 0.3644 yuan/kWh [43]. Assuming all generated electricity is sold to the grid, the centralized PV stations in the study area could potentially generate a revenue of 7.51 billion yuan.

4. Discussion

4.1. Comparison with existing research

The identification results were compared with existing publicly available PV datasets. Fig. 12(d)–(f) shows the PV distribution obtained by Feng et al. using Sentinel data and the PBIA method [25], and Fig. 12 (g)–(i) shows the PV distribution proposed by Wang et al. using Sentinel data and deep learning methods [37]. Compared to the PBIA results, the results of this study have more complete and smooth PV boundaries, especially for small, sporadically distributed PV areas. Relative to the deep learning results, the results of this study include more boundary details, with roads or valleys between PV areas accurately classified as no-PV areas. Furthermore, Fig. 12(i) displays a PV plant not identified by the deep learning method, which was successfully identified by our method with greater completeness than that identified by PBIA, shown in Fig. 12(m).

Table 9 shows the comparison of indicators between this study and previous studies. The results demonstrate that the proposed method achieves higher producer accuracy even in complex terrains, which is a critical factor in ensuring that real PV stations are not overlooked during the identification process. The test and training sets used in this study originate from different cities, presenting a more challenging scenario. Despite this, the method still maintains high overall accuracy.

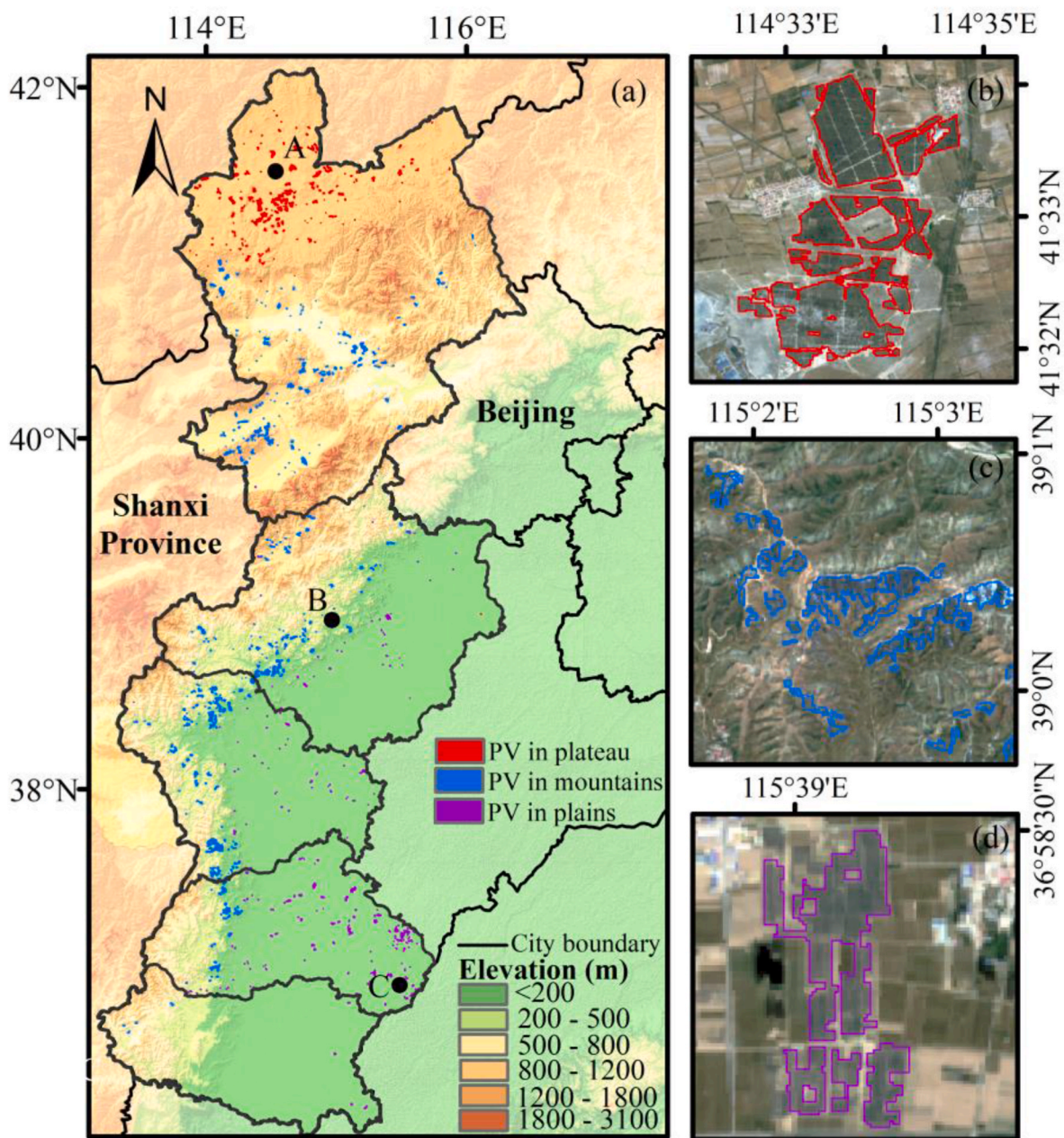


Fig. 9. Map of centralized PV distribution in five cities in western Hebei Province in 2024. (b) PV identification results in the plateau (c) PV identification results in the mountains (d) PV identification results in the plains.

Compared with PBIA, OBIA can effectively avoid the salt-and-pepper noise and does not need a complicated post-processing and manual check process. The PV results also have clearer and more complete boundaries with fewer missing pixels. Compared with existing deep learning methods, the present method identifies PV stations with more detailed and accurate boundaries, and significantly saves computational and storage resources for cutting and saving downloaded multi-band images.

Compared with the segmentation method of SNIC, multi-scale segmentation can automatically adjust the area and get fewer objects in a large homogeneous area, thus reducing the number of objects and improving the computational efficiency.

Compared with the research of Plakman et al. [31], our model outperforms their results in PA of PV and OA. Our study area is more complex than their plains study area. Their study area is flat, with PV stations, farmland, villages and other areas with nearly rectangular and

regular contours. In contrast, our study area encompasses diverse terrains, particularly smaller land-use patches with irregular shapes in mountainous regions. Despite these challenges, our UA of PV can still reach an approximate level with theirs.

Most models require a certain number of samples to be selected as a training set from every swath of application regions. This approach faces challenges that the region to be identified is located on an unfamiliar satellite swath. Expanding the training set by randomly changing the sample brightness consumes more computational resources [31]. In our method, this problem is alleviated by histogram matching and a second-level classifier.

4.2. Reliability of the PV stations mapping

With the global goal of carbon neutrality, PV has seen rapid growth in installed capacity in recent years. More and more PV stations are

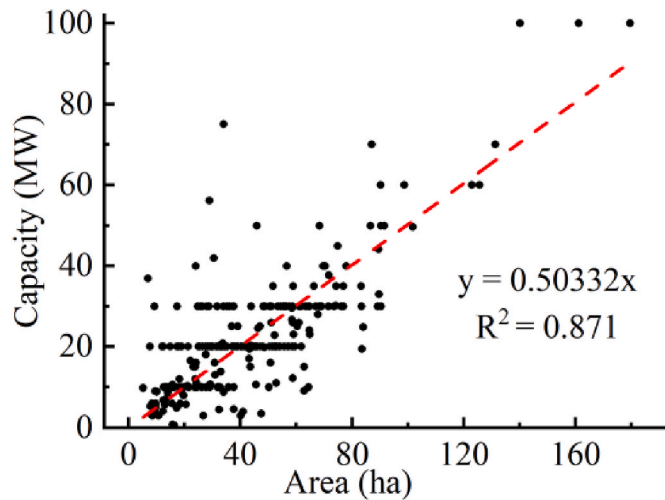


Fig. 10. Correlation between capacity and PV station area.

being installed in the mountains of China in order to be closer to electricity load centers and not to occupy the cropland. To accurately identify the area and distribution of centralized PV stations in complex terrains, we developed an RF cascade classifier model based on OBIA and Sentinel imagery. The model is able to accurately, comprehensively, and in detail identify fragmented PV areas in a new and unfamiliar region. This is the first time that centralized PV stations can be identified in the complete area without manual post-processing in a variety of terrain areas including mountains. It is also the first application of OBIA

in mountainous.

The success of this model and PV distribution map is attributed to the abundant Sentinel-1 and Sentinel-2 remote sensing data and color correction methods, appropriate OBIA image segmentation methods, reasonable segmentation parameters, optimal feature scenario, abundant and accurate labeled samples, the robust algorithm, and the cloud platform.

First, the 10m resolution sentinel image can well capture the fragmented PV in mountains, and Sentinel-1 images can overcome cloud coverage, making up for the Sentinel-2's shortcomings. In the study area, we obtained 274 Sentinel-2 images and 134 Sentinel-1 images to calculate the average. And the color correction with histogram matching is proved to be helpful to reduce the problem of spectral variability.

Second, the OBIA segments neighboring pixels with similar spectral features into a same object, and all pixels within each object are judged as a whole for the category. Compared with PBIA, the scale of the categorized objects is larger and the salt-and-pepper noise is effectively avoided. The image segmentation parameters were compared under various terrains, and the final applied segmentation parameters were able to fit the contours of the PV region under different terrains.

Third, 5 feature scenarios were compared, combined with the spectral, index, geometric, and texture features of Sentinel-1 and Sentinel-2. The index features, geometric features, and spectral features of Sentinel-1 have positive effects on the identification performance of the model. The introduction of Sentinel-1 spectral features significantly improves the performance. The area of objects is more important compared to other geometric features. However, the texture features, regardless of the window size, play a negative role in PV identification. This may be due to the large proportion of mountain PV stations in this study area, where

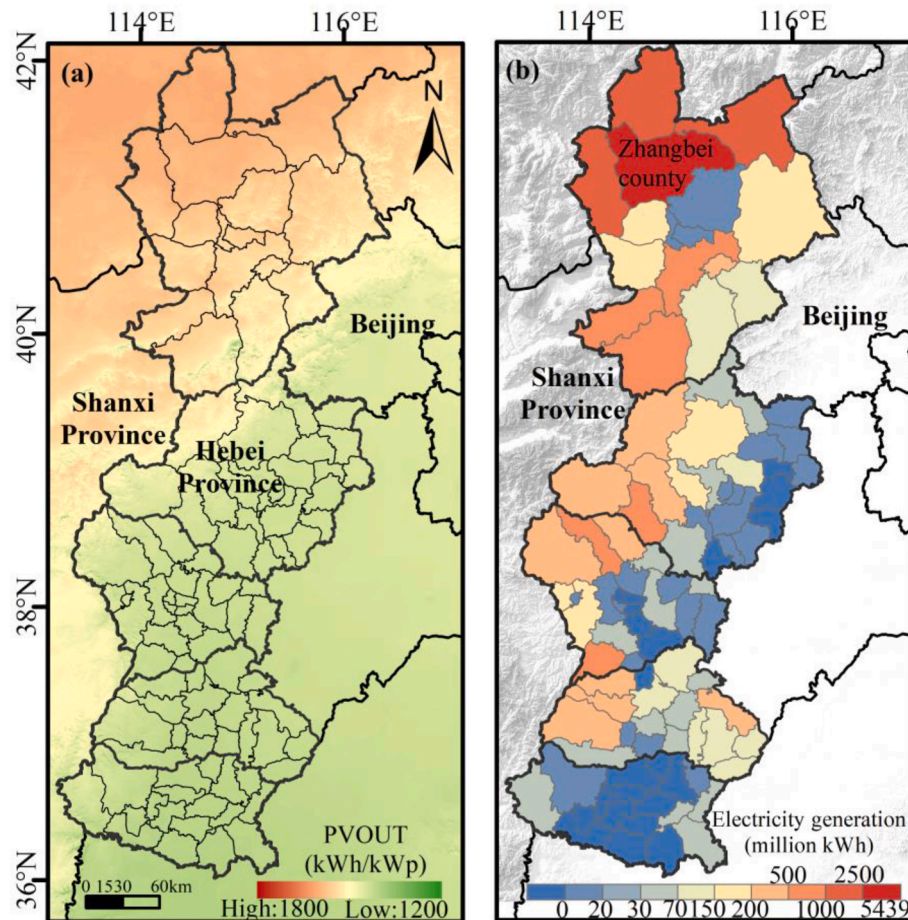


Fig. 11. Electricity generation estimation map. (a) PVOUT. (b) Annual electricity generation estimation of each county.

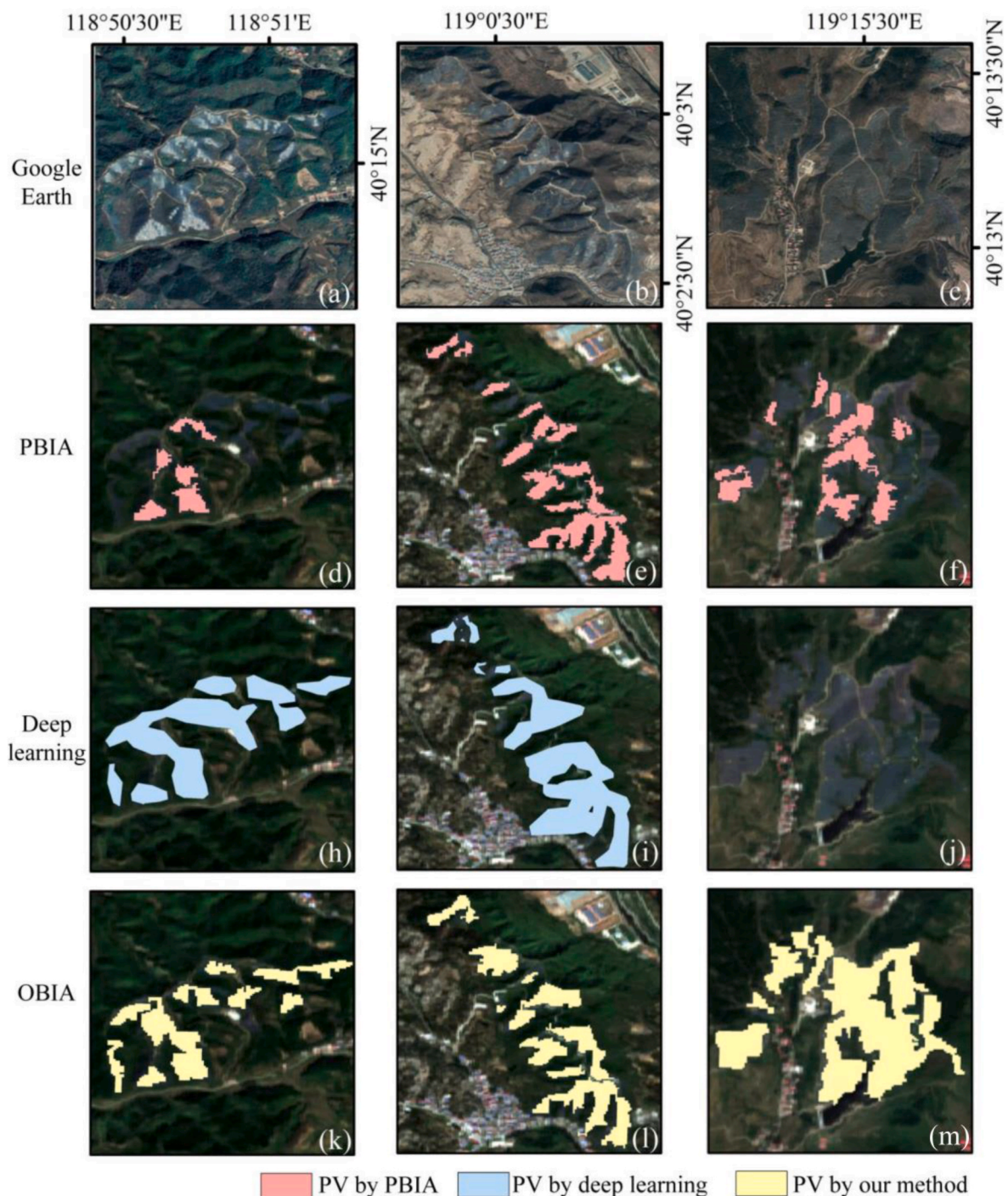


Fig. 12. Comparison of identification results of the method in this study with PBIA and deep learning on the test set. (a)~(c) Images with Ground truth. (d)~(f) Results by PBIA. (h)~(j) Results by deep learning. (k)~(m) Results by the method in this study. Each column represents the same PV station.

the orientation angle of some PV faces southwest or southeast, depending on the mountain orientation and resulting in atypical texture features. After streamlining features, the optimal first-level RF classifier contained 11 features, increasing computational efficiency.

Fourth, sufficient and accurate training samples are the basis for mapping [44]. Based on the existing public and available PV databases, we make full use of high-resolution Google Earth imagery to establish a large number of samples of centralized PV stations in different as a training set, with more accurate boundaries in the study area.

Last but not least, the study area in this study is one of the most difficult areas to identify in China due to the most complex topography

and many fragmented centralized PV in mountains. The use of RF cascade classifier has an excellent filtering effect in the complete regions, and it does not contain characteristic hyperparameters, making this method sufficiently robust in many regions.

4.3. Limitations and outlook

This paper presents the distribution map of PV stations in five cities of western Hebei Province for the year 2024. As PV stations continue to be constructed, the existing distribution data can be used to update the PV stations database in the future, utilizing newly available satellite

Table 9
Comparison of the accuracy metrics and statics of this study and previous studies.

Study	method	Image resolution	PA validation data	UA validation data	PA test data	UA test data	OA test data	Applied on unfamiliar areas	Terrain
Jiang et al. [24]	PBIA	Sentinel 1/2: 10m			88.70 %	98.20 %	96.90 %	No	Multiple
Xia et al. [30]	OBIA	Landsat: 30m	85.50 %	100 %				No	Plateaus
Plakman et al. [31]	OBIA	Sentinel 1/2: 10m	91.96 %	31.07 %	85.86 %	92.39 %	99.97 %	Yes	Plains
Wang et al. [37]	Deep learning	Sentinel 1/2: 10m	93.49 %	97.05 %			90.04 %	No	Multiple
This study	OBIA	Sentinel 1/2: 10m	95.52	96.12	92.73 %	81.38 %	99.99 %	Yes	Multiple

imagery.

Currently, the GEE platform does not incorporate the multi-scale segmentation algorithm, and can only download the images to local for segmentation. This process is complicated. It is found that if some PV stations in the region to be identified can be known and can be used as training set samples, the identification accuracy will be improved significantly. The increasing completeness of national PV station databases can facilitate more accurate identification.

5. Conclusion

Centralized PV stations are increasingly being installed in complex terrains, such as mountains and hills. The distribution of PV stations in these regions is fragmented, making it challenging for existing methods to generate comprehensive and accurate maps of PV distribution. This study proposes a method for extracting centralized PV stations at a 10-m resolution in complex terrains, using Sentinel-1/2 imagery, OBIA with multi-scale segmentation, and a RF cascade classifier. The method achieves optimal accuracy by utilizing a select combination of features from the Sentinel-2 bands, the Sentinel-1 VV band, the NDWI mean value, and the object area feature. In an independent, complete test region, the method achieves an overall accuracy of 99.9904 %, with a producer accuracy of 92.73 % and a user accuracy of 81.38 % for PV stations. A 2024 map of centralized PV stations was generated for five cities west of Beijing, revealing that PV installations in mountainous areas account for 45 % of the total installations in the region. This represents the first instance of identifying centralized PV stations across a complete region without the need for manual post-processing, in diverse terrains including mountains. Additionally, this is the first application of OBIA in both mountainous and plain terrains. Our method and the resulting data offer significant support for the statistical analysis and power generation predictions of centralized PV stations.

CRedit authorship contribution statement

Hongzhi Mao: Software, Methodology, Data curation, Conceptualization. **Weili Liu:** Resources, Investigation. **Xie Chen:** Resources, Investigation. **Zhiyong Tian:** Supervision, Funding acquisition. **Angelo Zarrella:** Writing – review & editing. **Yongqiang Luo:** Writing – review & editing. **Jianhua Fan:** Writing – review & editing. **Wentao Wu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Net Zero Tracker, Net zero numbers, Net Zero Tracker (2024). <https://zerotracker.net/>, 2024 8 May.

[2] National Energy Administration, Construction of Photovoltaic Power Generation in 2023 in China, National Energy Administration, 2024. https://www.nea.gov.cn/2024-02/28/c_1310765696.htm, 2024 28 February.

[3] Solar Energy Industries Association, U.S. Solar market insight 2023 Year in review, Solar Energy Ind. Assoc. (2024). <https://www.seia.org/us-solar-market-insight>, 2024 2024-10-15.

[4] Federal Network Agency, Statistics on Renewable Energy Sources, Federal Network Agency, 2024. <https://www.bundesnetzagentur.de/DE/Fachthemen/ElektrizitaetundGas/ErneuerbareEnergien/EE-Statistik/start.html>, 2024 13 May.

[5] Iea, World Energy Investment 2024, IEA, 2024. <https://www.iea.org/reports/world-energy-investment-2024>, 2024 June.

[6] European Commission, Renewable Energy Targets, European Commission, 2023. <https://energy.ec.europa.eu/topics/renewable-energy/renewable-energy-directive-targets-and-rules/>, 2024 15 May.

[7] Huiming Zhang, Zhidong Xu, Yi Zhou, et al., Optimal subsidy reduction strategies for photovoltaic poverty alleviation in China: a cost-benefit analysis, Resour. Conserv. Recycl. 166 (2021) 105352, <https://doi.org/10.1016/j.resconrec.2020.105352>.

[8] Huiming Zhang, Kai Wu, Yueming Qiu, et al., Solar photovoltaic interventions have reduced rural poverty in China, Nat. Commun. 11 (1) (2020) 1969, <https://doi.org/10.1038/s41467-020-15826-4>.

[9] Yongqiang Luo, Nan Cheng, Shicong Zhang, et al., Comprehensive energy, economic, environmental assessment of a building integrated photovoltaic-thermoelectric system with battery storage for net zero energy building, Build. Simulat. 15 (11) (2022) 1923–1941, <https://doi.org/10.1007/s12273-022-0904-1>.

[10] Mao Hongzhi, Xie Chen, Yongqiang Luo, et al., Advances and prospects on estimating solar photovoltaic installation capacity and potential based on satellite and aerial images, Renew. Sustain. Energy Rev. 179 (2023) 113276, <https://doi.org/10.1016/j.rser.2023.113276>.

[11] Xiaochun Li, Ze He, Siyou Xia, et al., Greenness change associated with construction and operation of photovoltaic solar energy in China, Renew. Energy 226 (2024) 120461, <https://doi.org/10.1016/j.renene.2024.120461>.

[12] Mohammad Masih Edalat, Haroon Stephen, Effects of two utility-scale solar energy plants on land-cover patterns using SMA of Thematic Mapper data, Renew. Sustain. Energy Rev. 67 (2017) 1139–1152, <https://doi.org/10.1016/j.rser.2016.09.079>.

[13] Zilong Xia, Yingjie Li, Wei Zhang, et al., Solar photovoltaic program helps turn deserts green in China: evidence from satellite monitoring, J. Environ. Manag. 324 (2022) 116338, <https://doi.org/10.1016/j.jenvman.2022.116338>.

[14] Hurdac Alexandra, Sofia L. Ermida, Miguel C. Brito, et al., Impact of a small-scale solar park on temperature and vegetation parameters obtained from Landsat 8, Renew. Energy 221 (2024) 119827, <https://doi.org/10.1016/j.renene.2023.119827>.

[15] Xie Chen, Hongzhi Mao, Nan Cheng, et al., Climate change impacts on global photovoltaic variability, Appl. Energy 374 (2024) 124087, <https://doi.org/10.1016/j.apenergy.2024.124087>.

[16] Qinglu Zeng, Luyao Li, Xie Chen, et al., Techno-economic performance and optimization of a large solar district heating system with pit storage under Tibetan Plateau climate conditions, Energy 315 (2025) 134449, <https://doi.org/10.1016/j.energy.2025.134449>.

[17] Tomohiro Ishii, Simo-Serra Edgar, Satoshi Iizuka, et al., Detection by classification of buildings in multispectral satellite imagery, in: 23rd International Conference on Pattern Recognition (ICPR), IEEE, Cancun, Mexico, 4–8 Dec. 2016, pp. 3344–3349, <https://doi.org/10.1109/ICPR.2016.7900150>, 2016.

[18] Xunhe Zhang, Mojtaba Zeraatpisheh, Md Mizanur Rahman, et al., Texture is important in improving the accuracy of mapping photovoltaic power plants: a case study of Ningxia autonomous region, China, Remote Sens. 13 (19) (2021) 3909, <https://doi.org/10.3390/rs13193909>.

- [19] X.H. Zhang, M. Xu, S.J. Wang, et al., Mapping photovoltaic power plants in China using Landsat, random forest, and Google Earth Engine, *Earth Syst. Sci. Data* 14 (8) (2022) 3743–3755, <https://doi.org/10.5194/essd-14-3743-2022>.
- [20] Z.H. Chen, Y.W. Kang, Z.X. Sun, et al., Extraction of photovoltaic plants using machine learning methods: a case study of the pilot energy city of Golmud, China, *Remote Sens.* 14 (11) (2022), <https://doi.org/10.3390/rs14112697>.
- [21] Xinxin Wang, Xiangming Xiao, Xi Zhang, et al., Characterization and mapping of photovoltaic solar power plants by Landsat imagery and random forest: a case study in Gansu Province, China, *J. Clean. Prod.* 417 (2023) 138015, <https://doi.org/10.1016/j.jclepro.2023.138015>.
- [22] L. Kruitwagen, K.T. Story, J. Friedrich, et al., A global inventory of photovoltaic solar energy generating units, *Nature* 598 (7882) (2021) 604–610, <https://doi.org/10.1038/s41586-021-03957-7>.
- [23] F. Ge, G.Z. Wang, G.J. He, et al., A hierarchical information extraction method for large-scale centralized photovoltaic power plants based on multi-source remote sensing images, *Remote Sens.* 14 (17) (2022), <https://doi.org/10.3390/rs14174211>.
- [24] Wenhao Jiang, Bo Tian, Yuanqiang Duan, et al., Rapid mapping and spatial analysis on the distribution of photovoltaic power stations with Sentinel-1&2 images in Chinese coastal provinces, *Int. J. Appl. Earth Obs. Geoinf.* 118 (2023) 103280, <https://doi.org/10.1016/j.jag.2023.103280>.
- [25] Quanlong Feng, Bowen Niu, Yan Ren, et al., A 10-m national-scale map of ground-mounted photovoltaic power stations in China of 2020, *Sci. Data* 11 (1) (2024) 198, <https://doi.org/10.1038/s41597-024-02994-x>.
- [26] Miguel-Ángel Manso-Callejo, Cira Calimanut-Ionut, Arranz-Justel José-Juan, et al., Assessment of the large-scale extraction of photovoltaic (PV) panels with a workflow based on artificial neural networks and algorithmic postprocessing of vectorization results, *Int. J. Appl. Earth Obs. Geoinf.* 125 (2023) 103563, <https://doi.org/10.1016/j.jag.2023.103563>.
- [27] Claudio Ladisa, Alessandra Capolupo, Maria Nicolina Ripa, et al., Combining OBIA approach and machine learning algorithm to extract photovoltaic panels from Sentinel-2 images automatically, in: *Remote Sensing for Agriculture, Ecosystems, and Hydrology XXIV*, 12262, SPIE, 2022, pp. 67–76, <https://doi.org/10.1117/12.2636451>.
- [28] Shiqi Tao, John Rogan, Su Ye, et al., Mapping photovoltaic power stations and assessing their environmental impacts from multi-sensor datasets in Massachusetts, United States, *Remote Sens. Appl.: Society and Environment* 30 (2023) 100937, <https://doi.org/10.1016/j.rsase.2023.100937>.
- [29] T. Stid Jacob, Siddharth Shukla, Annick Anctil, et al., Solar array placement, electricity generation, and cropland displacement across California's Central Valley, *Sci. Total Environ.* 835 (2022) 155240, <https://doi.org/10.1016/j.scitotenv.2022.155240>.
- [30] Zilong Xia, Yingjie Li, Ruishan Chen, et al., Mapping the rapid development of photovoltaic power stations in northwestern China using remote sensing, *Energy Rep.* 8 (2022) 4117–4127, <https://doi.org/10.1016/j.egy.2022.03.039>.
- [31] Veerle Plakman, Job Rosier, Jasper Van Vliet, Solar park detection from publicly available satellite imagery, *GIScience Remote Sens.* 59 (1) (2022) 461–480, <https://doi.org/10.1080/15481603.2022.2036056>.
- [32] Maximilian Schulz, Bilel Boughattas, Frank Wendel, DetEEktor: mask R-CNN based neural network for energy plant identification on aerial photographs, *Energy and AI* 5 (2021) 100069, <https://doi.org/10.1016/j.egyai.2021.100069>.
- [33] Chahine Bouaziz, El Koundi Mourad, Ghaleb Ennine, High-resolution solar panel detection in sfax, Tunisia: a UNet-based approach, *Renew. Energy* (2024) 121171, <https://doi.org/10.1016/j.renene.2024.121171>.
- [34] Yu Jiafan, Wang Zhecheng, Arun Majumdar, et al., DeepSolar: a machine learning framework to efficiently construct a solar deployment database in the United States, *Joule* 2 (12) (2018) 2605–2617, <https://doi.org/10.1016/j.joule.2018.11.021>.
- [35] Marek Vasku, An Exploration of Automatic Detection of Large-Scale Solar Plants: Application Of machine Learning-Based Image Classification in Google Earth Engine, Aalborg University, Copenhagen, Copenhagen, 2019.
- [36] Yongsheng Wang, Cui Xiao, Han Huang, Spatial patterns and environmental benefits of photovoltaic poverty alleviation programs in China, *Environ. Impact Assess. Rev.* 103 (2023) 107272, <https://doi.org/10.1016/j.eiar.2023.107272>.
- [37] Zirui Wang, Zhen Hao, Xiaofeng Jia, et al., Unravelling spatiotemporal patterns of solar photovoltaic plants development in China in the 21st century, *Environ. Res. Lett.* 19 (3) (2024) 034005, <https://doi.org/10.1088/1748-9326/ad25a3>.
- [38] Peiran Li, Haoran Zhang, Zhiling Guo, et al., Understanding rooftop PV panel semantic segmentation of satellite and aerial images for better using machine learning, *Adv. Appl. Energy* 4 (2021) 100057, <https://doi.org/10.1016/j.adapen.2021.100057>.
- [39] Compton J. Tucker, Red and photographic infrared linear combinations for monitoring vegetation, *Rem. Sens. Environ.* 8 (2) (1979) 127–150, [https://doi.org/10.1016/0034-4257\(79\)90013-0](https://doi.org/10.1016/0034-4257(79)90013-0).
- [40] S.K. Mcfeeters, The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features, *Int. J. Rem. Sens.* 17 (7) (1996) 1425–1432, <https://doi.org/10.1080/01431169608948714>.
- [41] Ministry of Finance Of China, Notice on Adjusting the Catalogue of Renewable Energy Price Additive Fund Subsidies (Photovoltaic Poverty Alleviation Projects), Ministry of Finance of China, 2020. https://www.gov.cn/zhengce/zhengceku/2020-08/26/content_5537487.htm, 2025 1 February.
- [42] Solargis, LT Aym Yearly Monthly Totals GEOTIFF PVOUT, Solargis, 2019. <https://solargis.com/cn/resources/free-maps-and-gis-data?locality=china>, 2025 1 February.
- [43] Polaris Solar Photovoltaic Network, Accelerating Market Entry: Photovoltaic Grid-Connected Prices in 31 Provinces, Regions, and Cities, Polaris solar photovoltaic network, 2024. <https://guangfu.bjx.com.cn/news/20240422/1372793.shtml>.
- [44] Álvaro Huerta Herraiz, Pliego Marugán Alberto, García Márquez, Fausto Pedro, Photovoltaic plant condition monitoring using thermal images analysis by convolutional neural network-based structure, *Renew. Energy* 153 (2020) 334–348, <https://doi.org/10.1016/j.renene.2020.01.148>.